EduConnect

Web Engineering Project

Containerization & Kubernetes Deployment

Complete Documentation

Frontend: React + Vite
Backend: Node.js + Express
Database: MongoDB (Replica Set)
Platform: Docker + Kubernetes (Minikube)

1. Project Overview

EduConnect is a full-stack tutoring platform that connects students with tutors. It supports three user roles — Student, Tutor, and Admin — each with dedicated dashboards, session management, verification workflows, and earnings tracking.

Component	Technology
Frontend	React 19 + Vite 8 + React Router
Backend	Node.js + Express + Nodemon
Database	MongoDB 6.0 (Replica Set)
Containerization	Docker
Orchestration	Kubernetes (Minikube)
Image Registry	Docker Hub (hashimaliii)

2. Project Structure

```
EduConnect-Web-Engineering/
├── client/                  # React frontend
│   ├── Dockerfile           # Multi-stage build (Node → nginx)
│   ├── .env                 # VITE_API_URL (baked at build time)
│   └── src/
├── server/                  # Express backend
│   ├── Dockerfile           # Node.js production image
│   └── server.js
├── infra/                   # Kubernetes manifests
│   ├── pv.yaml              # PersistentVolumes (3x for MongoDB)
│   ├── secret.yaml          # JWT secret
│   ├── mongo-ss.yaml        # MongoDB StatefulSet + headless Service
│   ├── server.yaml          # Backend Deployment + LoadBalancer Service
│   ├── client.yaml          # Frontend Deployment + LoadBalancer Service
│   └── hpa.yaml             # HorizontalPodAutoscalers
└── compose.yaml             # Docker Compose (local dev)
```

3. Docker Containerization

3.1 Server Dockerfile

The server image is a straightforward Node.js production image. It installs only production dependencies, switches to a non-root user for security, and starts via npm start (nodemon).

```
FROM node:lts-alpine
ENV NODE_ENV=production
WORKDIR /usr/src/app
COPY ["package.json", "package-lock.json*", "./"]
RUN npm install --production --silent && mv node_modules ../
COPY . .
EXPOSE 5000
RUN chown -R node /usr/src/app
USER node
CMD ["npm", "start"]
```

3.2 Client Dockerfile (Final — v3.0)

The original Dockerfile used npm run dev (Vite dev server) which crashes inside Kubernetes — it requires an interactive TTY and exits without one. The fix was a multi-stage build that compiles the React app into static files and serves them with nginx.

⚠ **Original Issue:** CMD ["npm", "run", "dev"] causes CrashLoopBackOff in Kubernetes because Vite's dev server requires a TTY and exits immediately in a headless environment.

```
# Stage 1: Build
FROM node:lts-alpine AS builder
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
ARG VITE_API_URL
ENV VITE_API_URL=$VITE_API_URL
RUN npm run build

# Stage 2: Serve with nginx
FROM nginx:alpine
COPY --from=builder /app/dist /usr/share/nginx/html
RUN echo 'server { listen 80; root /usr/share/nginx/html; \
  index index.html; location / { \
  try_files $uri $uri/ /index.html; } }' \
  > /etc/nginx/conf.d/default.conf
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

✔ **Why VITE_API_URL is a build ARG:** Vite bakes VITE_* variables into the JS bundle at build time. Passing it as a Docker ARG lets you embed the correct backend URL without modifying source files.

3.3 Building Images

```
# Server image
cd server
docker build -t hashimaliii/server:v1.0 .
docker push hashimaliii/server:v1.0

# Client image (v3.0 — production build with correct backend URL)
cd client
docker build --build-arg VITE_API_URL=http://127.0.0.1:5000 -t
hashimaliii/client:v3.0 .
docker push hashimaliii/client:v3.0
```

4. Docker Compose (Local Development)

Docker Compose is used for local development. It runs all three services — mongodb, server, and client — in a shared network with port mappings to the host.

```yaml
services:
  mongodb:
    image: mongodb/mongodb-community-server:latest
    container_name: mongodb
    restart: unless-stopped
    ports:
      - 27017:27017
    volumes:
      - mongodb:/data/db

  server:
    image: hashimaliii/server:v1.0
    env_file:
      - server/.env
    ports:
      - 5000:5000
    depends_on:
      - mongodb

  client:
    image: hashimaliii/client:v1.0
    env_file:
      - client/.env
    ports:
      - 5173:5173

volumes:
  mongodb:
    external: true
```

Running with Docker Compose

```bash
# Create the external volume first (only once)
docker volume create mongodb

# Start all services
docker compose up -d

# View logs
docker compose logs -f

# Stop
docker compose down
```

5. Kubernetes Deployment

5.1 Cluster Setup

```
# Start minikube
minikube start

# Enable metrics-server (required for HPA to read CPU metrics)
minikube addons enable metrics-server

# Verify storage class
kubectl get sc
# Should show: standard (default)   k8s.io/minikube-hostpath
```

5.2 Architecture

The Kubernetes deployment consists of the following resources:

Resource	Description
PersistentVolume (x3)	hostPath volumes on minikube node for MongoDB data
Secret	JWT secret key for the backend
StatefulSet (mongo)	3-pod MongoDB replica set with persistent storage
Headless Service (mongo)	DNS-based pod discovery for replica set members
Deployment (server)	3 replicas of the Node.js backend
Service (server-svc)	LoadBalancer — exposes backend on port 5000
Deployment (client)	3 replicas of the nginx frontend
Service (client-svc)	LoadBalancer — exposes frontend on port 80
HPA (server-hpa)	Autoscale server pods: min 2, max 5 at 70% CPU
HPA (client-hpa)	Autoscale client pods: min 2, max 5 at 70% CPU

5.3 Persistent Volumes (pv.yaml)

Three hostPath PersistentVolumes are pre-provisioned for MongoDB. Each maps to a directory on the minikube node. The StatefulSet's volumeClaimTemplates automatically binds to these.

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: mongo-pv-0
spec:
```

```
  capacity:
    storage: 1Gi
  accessModes:
    - ReadWriteOnce
  storageClassName: "standard"
  hostPath:
    path: /data/mongo-0
---
# mongo-pv-1 and mongo-pv-2 follow the same pattern
# with paths /data/mongo-1 and /data/mongo-2
```

5.4 Secret (secret.yaml)

```
apiVersion: v1
kind: Secret
metadata:
  name: server-secret
type: Opaque
stringData:
  jwt-secret: "supersecretkey123"
```

5.5 MongoDB StatefulSet (mongo-ss.yaml)

MongoDB is deployed as a StatefulSet to maintain stable network identities (mongo-0.mongo, mongo-1.mongo, mongo-2.mongo) needed for replica set configuration. A headless Service provides DNS-based discovery.

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: mongo
spec:
  serviceName: "mongo"
  replicas: 3
  selector:
    matchLabels:
      app: mongo
  template:
    metadata:
      labels:
        app: mongo
    spec:
      containers:
        - name: mongo
          image: mongo:6.0
          command: ["mongod", "--bind_ip_all", "--replSet", "rs0"]
          ports:
            - containerPort: 27017
          volumeMounts:
```

```
            - name: mongo-volume
              mountPath: /data/db
          resources:
            requests:
              cpu: "200m"
              memory: "256Mi"
            limits:
              cpu: "1000m"
              memory: "512Mi"
  volumeClaimTemplates:
    - metadata:
        name: mongo-volume
      spec:
        accessModes: ["ReadWriteOnce"]
        storageClassName: "standard"
        resources:
          requests:
            storage: 1Gi
---
apiVersion: v1
kind: Service
metadata:
  name: mongo
spec:
  clusterIP: None   # Headless — enables DNS pod discovery
  selector:
    app: mongo
  ports:
    - port: 27017
      targetPort: 27017
```

5.6 Server Deployment (server.yaml)

The backend Deployment runs 3 replicas. It reads JWT_SECRET from the Kubernetes Secret and connects to MongoDB via the replica set URI using pod DNS names.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: server
spec:
  replicas: 3
  selector:
    matchLabels:
      app: server
  template:
    spec:
      containers:
        - name: server
          image: hashimaliii/server:v1.0
          env:
            - name: MONGO_URI
```

```
              value: "mongodb://mongo-0.mongo:27017,mongo-1.mongo:27017,
                      mongo-2.mongo:27017/educonnect?replicaSet=rs0"
            - name: PORT
              value: "5000"
            - name: JWT_SECRET
              valueFrom:
                secretKeyRef:
                  name: server-secret
                  key: jwt-secret
          resources:
            requests:
              cpu: "100m"
              memory: "128Mi"
---
apiVersion: v1
kind: Service
metadata:
  name: server-svc
spec:
  type: LoadBalancer
  selector:
    app: server
  ports:
    - port: 5000
      targetPort: 5000
```

5.7 Client Deployment (client.yaml)

The frontend Deployment runs 3 replicas of the nginx image. The VITE_API_URL is baked into the image at build time, not passed as a runtime env var.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: client
spec:
  replicas: 3
  selector:
    matchLabels:
      app: client
  template:
    spec:
      containers:
        - name: client
          image: hashimaliii/client:v3.0
          ports:
            - containerPort: 80
          resources:
            requests:
              cpu: "100m"
              memory: "128Mi"
---
```

```yaml
apiVersion: v1
kind: Service
metadata:
  name: client-svc
spec:
  type: LoadBalancer
  selector:
    app: client
  ports:
    - port: 80
      targetPort: 80
```

5.8 Horizontal Pod Autoscaler (hpa.yaml)

HPA is configured for both client and server deployments. It scales pods between 2 and 5 replicas based on CPU utilization, targeting 70%.

```yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: server-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: server
  minReplicas: 2
  maxReplicas: 5
  metrics:
    - type: Resource
      resource:
        name: cpu
        target:
          type: Utilization
          averageUtilization: 70
---
# client-hpa follows the same spec targeting the client Deployment
```

6. Full Deployment Steps

Step 1 — Start Cluster

```
minikube start
minikube addons enable metrics-server
```

Step 2 — Apply Manifests in Order

```
kubectl apply -f infra/pv.yaml
kubectl apply -f infra/secret.yaml
kubectl apply -f infra/mongo-ss.yaml
kubectl apply -f infra/server.yaml
kubectl apply -f infra/client.yaml
kubectl apply -f infra/hpa.yaml
```

Step 3 — Wait for Pods

```
kubectl get pods -w
# Wait until all pods show: Running
```

Step 4 — Initialize MongoDB Replica Set

This step is required only once. It elects a PRIMARY and configures replication across the three mongo pods.

```
kubectl exec -it mongo-0 -- mongosh --eval "rs.initiate({
  _id: 'rs0',
  members: [
    { _id: 0, host: 'mongo-0.mongo:27017' },
    { _id: 1, host: 'mongo-1.mongo:27017' },
    { _id: 2, host: 'mongo-2.mongo:27017' }
  ]
})"

# Verify: should show 1 PRIMARY + 2 SECONDARY
kubectl exec -it mongo-0 -- mongosh --eval "rs.status()"
```

Step 5 — Expose Services (Windows / Docker Driver)

On Windows with the Docker driver, minikube tunnel is required to assign EXTERNAL-IP to LoadBalancer services. Open an Administrator PowerShell and keep it running:

```
# Administrator PowerShell — keep this window open
minikube tunnel

# In original terminal — verify EXTERNAL-IP is assigned
kubectl get services
# server-svc should show EXTERNAL-IP: 127.0.0.1
```

Step 6 — Build Client Image with Backend URL

```
cd client
docker build --build-arg VITE_API_URL=http://127.0.0.1:5000 -t
hashimaliii/client:v3.0 .
docker push hashimaliii/client:v3.0
cd ..
kubectl apply -f infra/client.yaml
```

Step 7 — Force Server Rollout (if pods are stale)

```
kubectl rollout restart deployment/server
kubectl logs deployment/server --follow
# Should show: Server running on port 5000 + MongoDB connected
```

Step 8 — Access the App

```
minikube service client-svc --url
# Opens the frontend in browser
```

7. Issues Encountered & Fixes

Issue	Root Cause & Fix
client pods: CrashLoopBackOff	Dockerfile used CMD [npm run dev]. Vite dev server exits without a TTY. Fixed by rewriting Dockerfile to multi-stage build with nginx.
npm run preview crash	No dist/ folder in image — no build step in original Dockerfile. Fixed by adding RUN npm run build in Stage 1.
VITE_API_URL = localhost:5000	Vite bakes env vars at build time. localhost does not resolve inside the cluster. Fixed by passing --build-arg VITE_API_URL at docker build time.
server-svc ClusterIP → no tunnel	ClusterIP is cluster-internal only. Browser cannot reach it. Fixed by changing service type to LoadBalancer and using minikube tunnel.
Service type cannot be changed in-place	kubectl apply on a service does not allow changing .spec.type. Fixed by kubectl delete service server-svc then kubectl apply.
MongoDB connection timeout	Server pods were stale (old env). kubectl rollout restart deployment/server forced new pods with correct MONGO_URI.
YAML parse errors in original infra files	Missing trailing newlines between multi-doc YAML files caused parse failures. Fixed by rewriting all infra files with proper separators.
HPA not working	metrics-server addon was not enabled. Fixed by minikube addons enable metrics-server. Also added resource requests to all deployments (required for HPA).

8. Verification Commands

```
# All pods running
kubectl get pods

# Services with EXTERNAL-IP
kubectl get services

# Persistent volumes and claims
kubectl get pv
kubectl get pvc

# HPA status
kubectl get hpa

# MongoDB replica set health
kubectl exec -it mongo-0 -- mongosh --eval "rs.status()"

# Server logs
kubectl logs deployment/server

# Describe a pod for events/errors
kubectl describe pod <pod-name>
```

9. Expected Final State

Resource	Expected State
client pods (x3)	Running — nginx serving React SPA
server pods (x3)	Running — Express API connected to MongoDB
mongo-0	Running — PRIMARY
mongo-1	Running — SECONDARY
mongo-2	Running — SECONDARY
client-svc	LoadBalancer — EXTERNAL-IP 127.0.0.1, port 80
server-svc	LoadBalancer — EXTERNAL-IP 127.0.0.1, port 5000
mongo (service)	Headless — ClusterIP None
server-hpa	min 2 / max 5 replicas, target CPU 70%
client-hpa	min 2 / max 5 replicas, target CPU 70%
PVC x3	Bound — 1Gi each for mongo-0, mongo-1, mongo-2

Name	Tag	Image ID	Created	Size	Actions
hashimaliii/client	v1.0	6eafcb9d3f67	1 day ago	463.18 MB	
client	latest	6eafcb9d3f67	1 day ago	463.18 MB	
hashimaliii/server	v1.0	8c0c35392f49	2 days ago	262.47 MB	
server	latest	8c0c35392f49	2 days ago	262.47 MB	
mongodb/mongodb-community-se	latest	4dd45ef5bace	2 days ago	1.2 GB	
docker.n8n.io/n8nio/n8n	latest	342eeafdb289	9 days ago	1.98 GB	
gcr.io/k8s-minikube/kicbase	v0.0.50	b97074569ae9	29 days ago	1.93 GB	
gcr.io/k8s-minikube/kicbase	<none>	eb4fec00e8ad	29 days ago	1.93 GB	
atendai/evolution-api	v1.8.2	1534d64a128f	2 years ago	1.02 GB	
hashimaliii/client	v2.0	a6e229efcf10	33 minutes ag	93.55 MB	
hashimaliii/client	v3.0	172e9d7284f6	23 minutes ag	93.55 MB	

docker desktop PERSONAL

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Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions

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View configurations

server
hashimaliii/server:v1.0
5000:5000

client
hashimaliii/client:v1.0
5173:5173

mongodb
mongodb/mongodb-cor
27017:27017

```
:1000000,"category_id":7,"verbose_level":"INFO","verbose_level_id":0,"msg":"saving checkpoint snapshot min: 184, snapshot max: 184 snapshot count: 0,
oldest timestamp: (0, 0) , meta checkpoint timestamp: (0, 0) base write gen: 1929"}}}
{"t":{"$date":"2026-03-19T14:58:18.943+00:00"},"s":"I",  "c":"WTCHKPT",  "id":22430,   "ctx":"Checkpointer","msg":"WiredTiger message","attr":{"message":
{"ts_sec":1773932298,"ts_usec":943516,"thread":"8:0x725728249640","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","log_id"
:1000000,"category_id":7,"verbose_level":"INFO","verbose_level_id":0,"msg":"saving checkpoint snapshot min: 185, snapshot max: 185 snapshot count: 0,
oldest timestamp: (0, 0) , meta checkpoint timestamp: (0, 0) base write gen: 1929"}}}
{"t":{"$date":"2026-03-19T14:59:18.974+00:00"},"s":"I",  "c":"WTCHKPT",  "id":22430,   "ctx":"Checkpointer","msg":"WiredTiger message","attr":{"message":
{"ts_sec":1773932358,"ts_usec":974611,"thread":"8:0x725728249640","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","log_id"
:1000000,"category_id":7,"verbose_level":"INFO","verbose_level_id":0,"msg":"saving checkpoint snapshot min: 186, snapshot max: 186 snapshot count: 0,
oldest timestamp: (0, 0) , meta checkpoint timestamp: (0, 0) base write gen: 1929"}}}
{"t":{"$date":"2026-03-19T15:00:19.020+00:00"},"s":"I",  "c":"WTCHKPT",  "id":22430,   "ctx":"Checkpointer","msg":"WiredTiger message","attr":{"message":
{"ts_sec":1773932419,"ts_usec":20136,"thread":"8:0x725728249640","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","log_id":
1000000,"category_id":7,"verbose_level":"INFO","verbose_level_id":0,"msg":"saving checkpoint snapshot min: 187, snapshot max: 187 snapshot count: 0,
oldest timestamp: (0, 0) , meta checkpoint timestamp: (0, 0) base write gen: 1929"}}}
{"t":{"$date":"2026-03-19T15:01:19.045+00:00"},"s":"I",  "c":"WTCHKPT",  "id":22430,   "ctx":"Checkpointer","msg":"WiredTiger message","attr":{"message":
{"ts_sec":1773932479,"ts_usec":45448,"thread":"8:0x725728249640","session_name":"WT_SESSION.checkpoint","category":"WT_VERB_CHECKPOINT_PROGRESS","log_id":
1000000,"category_id":7,"verbose_level":"INFO","verbose_level_id":0,"msg":"saving checkpoint snapshot min: 188, snapshot max: 188 snapshot count: 0,
oldest timestamp: (0, 0) , meta checkpoint timestamp: (0, 0) base write gen: 1929"}}}
```

client

```
> client@0.0.0 dev
> vite


  VITE v8.0.1  ready in 893 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: http://172.19.0.3:5173/
```

server

```
> server@1.0.0 start
> nodemon server.js

[nodemon] 3.1.9
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Server running on port 5000
```

mongodb

```
{"t":{"$date":"2026-03-20T10:17:59.663+00:00"},"s":"I",  "c":"NETWORK",  "id":22943,   "ctx":"listener","msg":"Connection accepted","attr":
{"remote":"172.19.0.4:45804","isLoadBalanced":false,"uuid":{"$uuid":"c99e06e0-d462-499d-a602-
f624d7c64127"}},"connectionId":1,"connectionCount":1}}
```

Engine running | | RAM 5.86 GB CPU 7.19% Disk: 14.79 GB used (limit 1006.85 GB) Terminal Update available

```
PROBLEMS 17   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   GITLENS   SPELL CHECKER 37

PS D:\Projects\EduConnect-Web-Engineering> # 1. Running Docker containers (from earlier Docker Compose run)
>> docker ps
>>
>> # 2. Running Kubernetes pods
>> kubectl get pods
>>
>> # 3. Services
>> kubectl get services
>>
>> # 4. Scaling / HPA
>> kubectl get hpa
>>
>> # 5. Persistent volumes
>> kubectl get pv
>> kubectl get pvc
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS         PORTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
NAMES
b6dad0b1369f   gcr.io/k8s-minikube/kicbase:v0.0.50       "/usr/local/bin/entr…"   14 hours ago   Up 42 minutes   127.0.0.1:57046->22/tcp, 127.0.0.1:57047->2376/tcp, 127.0.0.1:57044->5000/tcp, 127.0.0.1:57045->8443/tcp, 127.0.0.1:57043->32443/tcp   m
inikube
55dee77750c8   mongodb/mongodb-community-server:latest   "python3 /usr/local/…"   22 hours ago   Up 43 minutes   0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
ongodb
47b9696894ba   docker.n8n.io/n8nio/n8n                  "tini -- /docker-ent…"   7 days ago     Up 43 minutes   0.0.0.0:5678->5678/tcp, [::]:5678->5678/tcp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
8n_brain
b1739d78a4c6   atendai/evolution-api:v1.8.2             "docker-entrypoint.s…"   7 days ago     Up 43 minutes   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
volution_api
NAME                      READY   STATUS    RESTARTS   AGE
client-5b7b8588c-8j1p7    1/1     Running   0          21m
client-5b7b8588c-kn8rc    1/1     Running   0          21m
mongo-0                   1/1     Running   0          40m
mongo-1                   1/1     Running   0          40m
mongo-2                   1/1     Running   0          40m
server-99855bb58-k22wt    1/1     Running   0          2m
server-99855bb58-nthzx    1/1     Running   0          2m4s
NAME         TYPE           CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
client-svc   LoadBalancer   10.111.180.137 127.0.0.1     80:31382/TCP     12h
kubernetes   ClusterIP      10.96.0.1      <none>        443/TCP          14h
mongo        ClusterIP      None           <none>        27017/TCP        13h
server-svc   LoadBalancer   10.106.28.255  127.0.0.1     5000:30732/TCP   22m
NAME         REFERENCE           TARGETS           MINPODS   MAXPODS   REPLICAS   AGE
client-hpa   Deployment/client   cpu: 0%/70%       2         5         2          40m
server-hpa   Deployment/server   cpu: <unknown>/70%  2         5         2          40m
NAME         CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM                          STORAGECLASS   VOLUMEATTRIBUTESCLASS   REASON   AGE
mongo-pv-0   1Gi        RWO            Retain           Bound    default/mongo-volume-mongo-1   standard       <unset>                          41m
mongo-pv-1   1Gi        RWO            Retain           Bound    default/mongo-volume-mongo-2   standard       <unset>                          41m
mongo-pv-2   1Gi        RWO            Retain           Bound    default/mongo-volume-mongo-0   standard       <unset>                          41m
NAME                   STATUS   VOLUME       CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mongo-volume-mongo-0   Bound    mongo-pv-2   1Gi        RWO            standard       <unset>                 40m
mongo-volume-mongo-1   Bound    mongo-pv-0   1Gi        RWO            standard       <unset>                 40m
mongo-volume-mongo-2   Bound    mongo-pv-1   1Gi        RWO            standard       <unset>                 40m
PS D:\Projects\EduConnect-Web-Engineering>
```